Portfolio Optimization: A Monte Carlo Study

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Problem Definition: Four Risky Assets

Consider a portfolio allocation problem involving four risky assets with expected annual returns μ_i, volatilities (standard deviations) σ_i, and a dependence structure (correlation matrix) as follows:

Asset	μ	σ	$\mathbf{w}$
A	0.02	0.05	w_1
B	0.07	0.12	w_2
C	0.15	0.17	w_3
D	0.20	0.25	w_4

$$\mathbf{R} = \begin{pmatrix} 1 & 0.3 & 0.3 & 0.3 \\ 0.3 & 1 & 0.6 & 0.6 \\ 0.3 & 0.6 & 1 & 0.6 \\ 0.3 & 0.6 & 0.6 & 1 \end{pmatrix}$$

The corresponding portfolio optimization problem involves solving for the portfolio weights $\mathbf{w}$, given investment constraints.

Rather than solving this constrained optimization problem with calculus (using Lagrange multipliers) or with mathematical programming (using a quadratic objective function), we explore the domain of possible portfolio allocations in a Monte Carlo study.

Monte Carlo Methods

Here are the steps involved in the Monte Carlo study:

- Generate above 700 random allocation sets: 4×1 vectors. Each set has to satisfy the constraint $w^T \mathbf{1} = 1$. In fact, once we generate three random numbers w_1, w_2, w_3, the 4th can be computed.
- Compute the portfolio return $\mu_\Pi = \mathbf{w^T}\mu$ and volatility $\sigma_\Pi = \sqrt{\mathbf{w^T \Sigma w}}$ for each set $\mathbf{w}$.
- Plot points with coordinates μ_Π on the vertical axis and σ_Π on the horizontal axis.
- Note that many sets of weights will not be optimal— portfolio returns can be negative.

Regarding investment constraints, we consider two asset management strategies: (1) a conservative approach that takes long positions only and (2) a more flexible approach that takes both long and short positions.

We work with asset means, volatilities (standard deviations), and correlations as defined in the four risky asset problem. We compute the target asset covariance matrix from the standard deviations and correlations, and we generate 700 asset return sets assuming a multivariate normal distribution.

To test asset management strategies, we generate 700 random asset weightings under a long-positions-only strategy and another 700 random asset weightings under a more flexible longs-and-shorts strategy.

The Monte Carlo study was implemented in R using the MASS library. See the program file: **451-portfolio-optimization-monte-carlo.R**.

Results and Conclusions

Figure 1. shows the contrasting opportunity sets for the two investment strategies. Portfolios with all positive weights (no shorts) are shown in blue. Portfolios with at least one short position are shown in red.

The long-positions-only strategy depicted in the left-hand panel of figure 1, while less risky overall, limits the opportunity set to portfolios earning less than a 0.20 return. The long-positions-only strategy is risk-adverse. It advoids portfolios with negative returns.

To achieve higher returns (indeed, returns above 0.40), an asset manager must be open to the possibility of taking short positions, as depicted in the right-hand panel of figure 1.

The left-hand edges above 0 return on the vertial axis show the efficient frontiers associated with the two investment strategies.

A key takeaway from this study for traders and asset managers:

> **If you have a moderate level of risk tolerance, to ensure higher returns, you should be open to the possibility of taking short positions.**

Figure 1: Opportunity Sets of Risky Assets and Portfolios

